

8 – BIT MICROPROCESSOR (8085): FEATURES AND ARCHITECTURE

Features:

- Intel 8085 is an 8-bit, NMOS microprocessor.
- It is a 40 pin I.C. package fabricated on a single LSI chip.
- The Intel 8085 uses a single +5Vd.c. supply for its operation.
- Its clock speed is about 3 MHz
- It is enclosed with 40 pins DIP (Dual in line package)
- The clock cycle is of 320 ns.
- The width of the data bus is 8 – bit and for address bus is 16- bits and hence can address up to $2^{16} = 65536$ bytes (64KB) memory locations.
- It supports external interrupt request.

8085 MICROPROCESSOR ARCHITECTURE

The 8085 microprocessor is an 8-bit processor available as a 40-pin IC package and uses +5 V for power. It can run at a maximum frequency of 3 MHz Its data bus width is 8-bit and address bus width are 16-bit, thus it can address $2^{16} = 64$ KB of memory. The internal architecture of 8085 is shown in Fig.

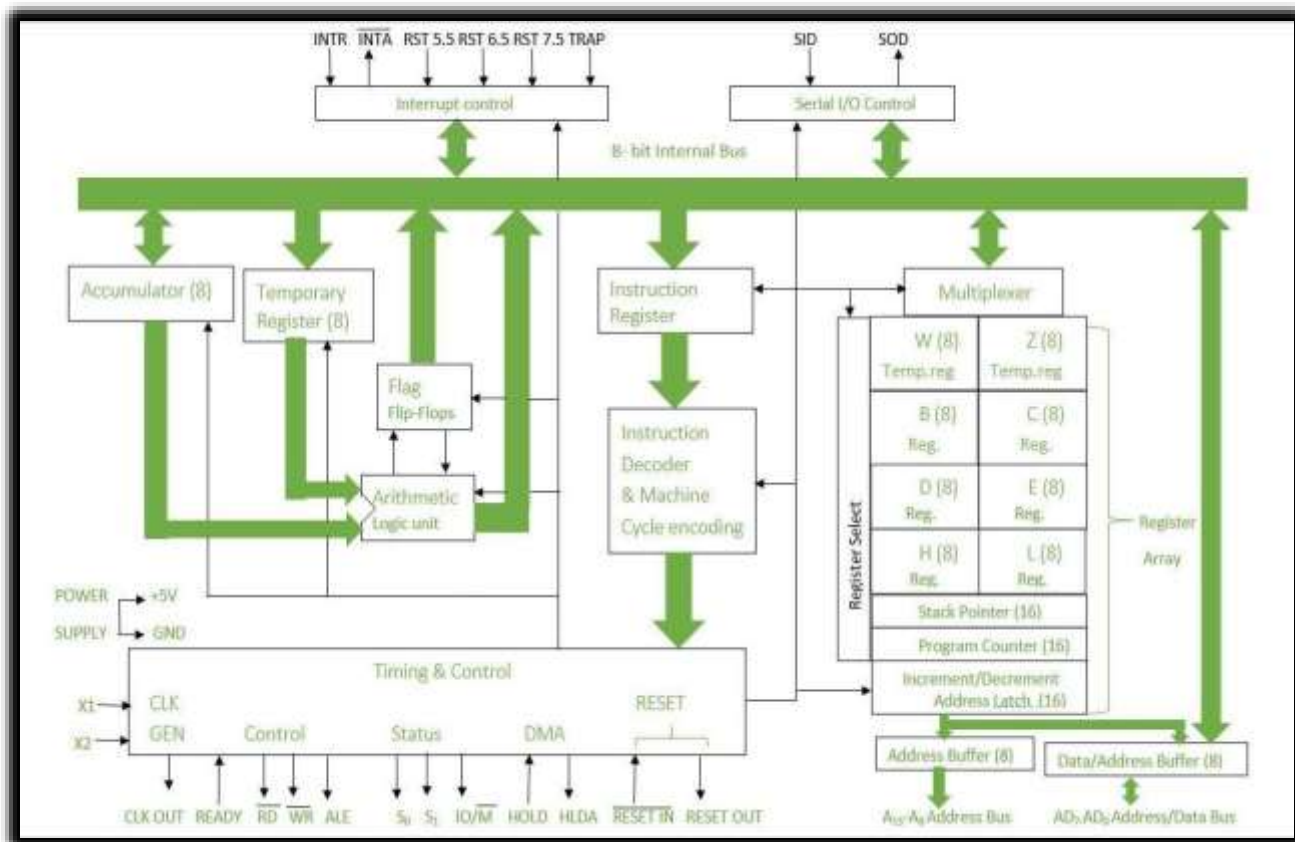


Fig. Architecture of 8085

It consists of three main sections:

- [1] Arithmetic and Logic Unit (ALU)
- [2] Timing & Control Unit
- [3] Set of registers.

1. Arithmetic and Logic Unit

The ALU performs the actual numerical and logical operations such as

- Addition (ADD)
- Subtraction (SUB)
- Logical AND
- Logical OR
- Logical EX-OR.
- Increment (add 1)
- Decrement (Subtract 1)
- Left shift, Rotate left, Rotate Right.

2. Timing & Control Unit:

- It generates timing & control signals which are necessary for the execution of instructions.
- It controls data flow between CPU and peripherals including memory.
- It provides status, control and timing signals which are required for the operation of memory and I/O devices.
- It controls entire operations of the microprocessor and peripherals connected to it.
- Thus, it is seen that the control unit of the CPU acts as the brain of the computer system.

3. Registers:

Intel 8085 microprocessor has the following registers:

- i. One 8-bit Accumulator (ACC) i.e. register A
- ii. Six 8-bit general purpose registers. These are B, C, D, E, H & L
- iii. One 16-bit stack pointer, SP
- iv. One 16-bit program Counter, PC
- v. Instruction registers
- vi. Temporary register
- vii. Flag register

Accumulator (ACC):

- The accumulator (register A) is an 8-bit register associated with ALU.
- It is used to hold one of the operands of an arithmetic or logical operation.
- It serves as one input to the ALU. The other operand for an arithmetic or logical operation may be stored either in the memory or in one of the general-purpose registers.
- Final result of an arithmetic or logical operation is placed in the accumulator.

General Purpose Registers

- The 8085 microprocessor contains six 8-bit general purpose registers. They are B, C, D, E, H and L.
- To hold 16-bit data a combination of two 8-bit registers can be employed which known as register pairs.
- The valid register pairs in 8051 are B-C, D-E and H-L. the programmer cannot form the register pair by selecting any two registers of his choice.
- The H-L pair is used to act as memory pointer and for this purpose it holds the 16-bit address of a memory location.
- The general-purpose registers and the accumulator are accessible to programmer. He can store data in these registers during writing his program.

Program Counter (PC):

- It is a 16-bit special purpose register used to hold the memory address of the next instruction which is to be executed.
- It keeps the track of memory addresses of the instructions in the program while they are being executed.
- The microprocessor increments the content of the program counter during the execution of an instruction so that it points to the address of the next instruction in the program at the end of the execution of an instruction.

Stack Pointer (SP):

- It is a 16-bit special purpose register.
- Stack is a sequence of memory location set aside by the programmer to store/retrieve the content of accumulator, flags, program counter and general-purpose registers during the execution of a program.
- Any portion of the memory can be used as stack. Since the stack works on LIFO (last in first out) principle, its operation is faster compared normal store/ retrieve of memory locations.

- The SP holds the address of the top element of data stored in the stack.
- The stack is defined and stack pointer is initialized by the programmer at the beginning of a program which needs stack operation. Stack is also used by the microprocessor. For example, it stores the contents of program counter when it jumps to a subroutine using CALL instruction.

Instruction Register:

- The instruction register holds the opcode (operation code or instruction code) of the instruction which is being decoded and executed.

Temporary Register:

- It is an 8-bit register associated with ALU. It holds data during an arithmetic/logical operation.
- It is used by microprocessor.
- It is not accessible to programmer.

Flags (Program Status Word i.e. PSW):

- The Intel 8085 microprocessor contains five flip-flops to serve as status flags.
- The flip-flops are set or reset according to the conditions which arises during an arithmetic or logical operation.
- The five status flags of Intel 8085 are:
 - Carry flag (CS)
 - Parity Flag (P)
 - Auxiliary Carry Flag (AC)
 - Zero Flag (Z)
 - Sign Flag (S)

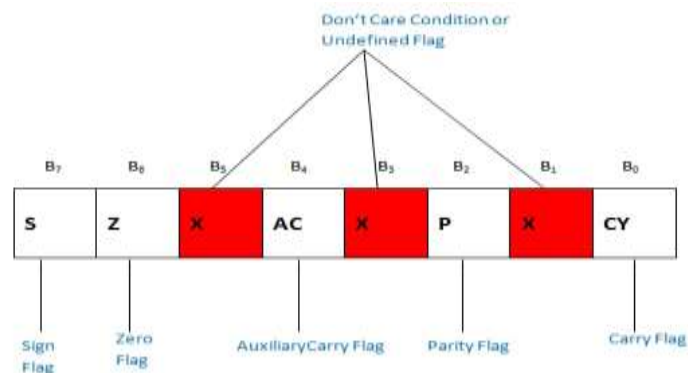


Fig. Flag Register

In fig above, five bits indicates the five status flags and three bits are undefined. “The combination of these 8-bits is called Program Status Word (PSW)”. PSW and the accumulator is treated as a 16-bit unit for stack operation.

Carry Flag (CS):

- After execution of an arithmetic instruction if carry is produced, the carry flag CS is set to 1. Otherwise it is 0.
- The carry flag is set or reset in case of addition as well as subtraction.
- After the addition of two 8-bit numbers, if the sum is larger than 8-bits, a carry is producing; and the carry flag is set to 1.
- In case of subtraction, if borrow occurs, the carry flag is set to 1.
- The carry flag holds carry out of the most significant bit resulting from the execution of an arithmetic operation.

Parity Flag (P):

- The parity flag P is set to 1, if the result of an arithmetic or logical operation contains even number of 1s.
- It is reset i.e. it is 0, if the result contains odd number of 1s.

Auxiliary Carry Flag (AC):

- The auxiliary carry flag AC hold carry out of the bit number 3 to the bit number 4 resulting from the execution of an arithmetic operation.
- The counting of bits starts from 0, and hence Bit No.3 is actually the fourth bit from the least significant bit.

Zero Flag (Z):

The zero-status flag Z is set to 1, if the result of an arithmetic or logical operation is Zero otherwise it is set to 0.

Sign Flag (S):

- The sign flag S is set to 1, if the result of an arithmetic or logical operation is negative. If the result is positive, the sign flag is set to 0.
- The MSB is used as a sign bit.

Serial Input/Output Control

It controls the serial data communication by using Serial input data and Serial output data.

Serial Input/Output control in the 8085 microprocessor refers to the communication of data between the microprocessor and external devices in a serial manner, i.e., one bit at a time.

The 8085 has a serial I/O port (SID/SOD) for serial communication. The SID pin is used for serial input and the SOD pin is used for serial output. The timing and control of serial communication is managed by the 8085's internal circuitry.

Data and Address bus:

- The Intel 8085 is an 8-bit microprocessor. Its data bus is 8-bit wide and hence, 8bit of data can be transmitted in parallel from or to the microprocessor.
- The Intel 8085 requires a 16-bit wide address bus as the memory addresses are of 16 bits.
- The 8 most significant bits of the address are transmitted by the address bus, a-bus (pins A8 to A15).
- The least significant bits of address are transmitted by address/ data bus, AD-bus (pins AD0 to AD7).
- The higher order address lines (A8 – A15) are unidirectional and the lower order lines (A0 – A7) are multiplexed (time-shared) with the eight data bits (D0 – D7) and hence, they are bidirectional.
- The address/ data bus transmits data and address at different moments. At a particular moment it transmits either data or address. Thus AD-bus operates in time shared mode.

Control Bus:

Control bus is various lines which have specific functions for coordinating and controlling microprocessor operations. The control bus carries control signals partly unidirectional and partly bidirectional. The following control and status signals are used by 8085 processor:

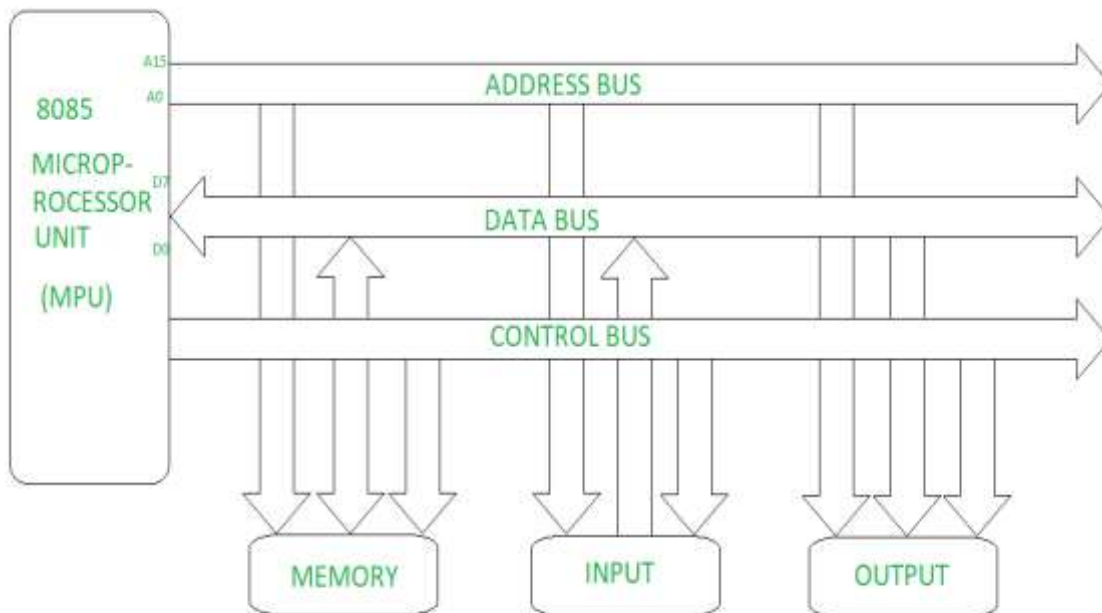
- ALE (output): Address Latch Enable is a pulse that is provided when an address appears on the AD0 – AD7 lines, after which it becomes 0.
- $\overline{RD}$ (active low output): The Read signal indicates that data are being read from the selected I/O or memory device and that they are available on the data bus.
- $\overline{WR}$ (active low output): The Write signal indicates that data on the data bus are to be written into a selected memory or I/O location.
- $\overline{IO/M}$ (output): It is a signal that distinguished between a memory operation and an I/O operation. When $\overline{IO/M}=0$ it is a memory operation and $\overline{IO/M}=1$ it is an I/O operation.

- S1 and S0 (output): These are status signals used to specify the type of operation being performed; they are listed in Table 1.

Table 1 Status signals and associated operations

S1	S0	States
0	0	Halt
0	1	Write
1	0	Read
1	1	Fetch

The schematic representation of the 8085-bus structure is as shown in Fig.



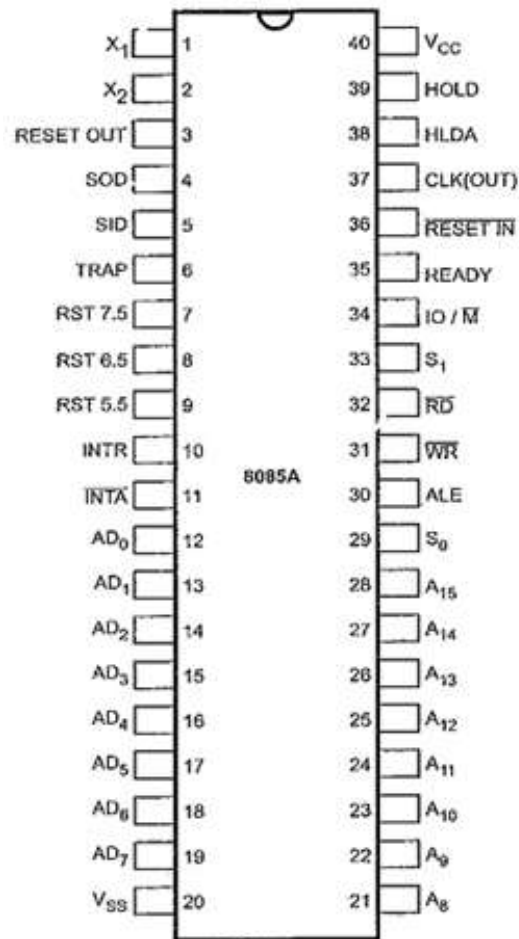
Bus organization system of 8085 Microprocessor

The microprocessor performs primarily four operations:

- I. Memory Read: Reads data (or instruction) from memory.
- II. Memory Write: Writes data (or instruction) in to memory.
- III. I/O Read: Accepts data from input device.
- IV. I/O Write: Sends data to output device.

PIN CONFIGURATION OF 8085 MICROPROCESSOR

The pin configuration of 8085 microprocessor is shown below.



It consists of,

- Power Supply and Frequency Signals
- Data Bus and Address Bus
- Control and Status Signals
- Interrupt Signals
- Serial I/O Signals
- DMA Signal
- Reset Signals

1. Power Supply and Frequency Signals:

- V_{CC} : It requires a single +5 V power supply.
- V_{SS} : Ground reference.

- X_1 and X_2 : A tuned circuit like LC, RC or crystal is connected at these two. The internal clock generator divides oscillator frequency by 2, therefore, to operate a system at 3 MHz, the crystal of tuned circuit must have a frequency of 6 MHz.
- CLK OUT : This signal is used as a system clock for other devices. Its frequency is half the oscillator frequency.

2. Data Bus and Address Bus:

A) AD_0 to AD_7 : The 8 bit data bus ($D_0 - D_7$) is multiplexed with the lower half ($A_0 - A_7$) of the 16 bit address bus. During first part of the machine cycle (T_1), lower 8 bits of memory address or I/O address appear on the bus. During remaining part of the machine cycle (T_2 and T_3) these lines are used as a bi-directional data bus.

B) A_8 to A_{15} : The upper half of the 16-bit address appears on the address lines A_8 to A_{15} . These lines are exclusively used for the most significant 8 bits of the 16-bit address lines.

3. Control and Status Signals:

A) ALE (Address Latch Enable) : AD_0 to AD_7 lines are multiplexed and the lower half of address ($A_0 - A_7$) is available only during T_1 of the machine cycle. This lower half of address is also necessary during T_2 and T_3 of machine cycle to access specific location in memory or I/O port. This means that the lower half of an address must be latched in T_1 of the machine cycle, so that it is available throughout the machine cycle. The latching of lower half of an address bus is done by using external latch and ALE signal from 8085 Pin Diagram.

B) RD and WR : These signals are basically used to control the direction of the data flow between processor and memory or I/O device/port. A low on RD indicates that the data must be read from the selected memory location or I/O port via data bus. A low on WR indicates that the data must be written into the selected memory location or I/O port via data bus.

C) IO/M, S_0 and S_1 : IO/M indicates whether I/O operation or memory operation is being carried out. S_1 and S_0 indicate the type of machine cycle in progress.

D) READY: It is used by the microprocessor to sense whether a peripheral is ready or not for data transfer. If not, the processor waits. It is thus used to synchronize slower peripherals to the microprocessor.

4. Interrupt Signals:

The 8085 Pin Diagram has five hardware interrupt signals: RST 5.5, RST 6.5, RST 7.5, TRAP and INTR. The microprocessor recognizes interrupt requests on these lines at the end of the current instruction execution.

The INTA (Interrupt Acknowledge) signal is used to indicate that the processor has acknowledged an INTR interrupt.

5. Serial I/O Signals:

A) SID (Serial I/P Data): This input signal is used to accept serial data bit by bit from the external device.

B) SOD (Serial O/P Data) : This is an output signal which enables the transmission of serial data bit by bit to the external device.

6. DMA Signal:

A) HOLD: This signal indicates that another master is requesting for the use of address bus, data bus and control bus.

B) HLDA: This active high signal is used to acknowledge HOLD request.

7. Reset Signals:

A) RESET IN: A low on this pin

- Sets the program counter to zero (0000H).
- Resets the interrupt enable and HLDA flip-flops.
- Tri-states the data bus, address bus and control bus.
- Affects the contents of processor's internal registers randomly.

On reset, the PC sets to 0000H which causes the 8085 Pin Diagram to execute the first instruction from address 0000H. For proper reset operation reset signal must be held low for at least 3 clock cycles. The power-on reset circuit can be used to ensure execution of first instruction from address 0000H.

B) RESET OUT: This active high signal indicates that processor is being reset. This signal is synchronized to the processor clock and it can be used to reset other devices connected in the system.